

PREOPERATIVE ASSESSMENT & GENERAL ANAESTHESIA

Preoperative evaluation

- Aims: ensure safety, optimize condition, minimize risk. Begins when anaesthetist reviews record + visits patient day before elective surgery.
- **Includes:** history, current drugs, exam, labs, **consent**, anaesthesia plan, postop pain relief, fasting, **ASA classification**, premedication.

ASA Fasting guidelines

Material	Minimum fast
Clear liquids	**2 h**
Breast milk	4 h
Infant formula	5–6 h
Non-human milk	6 h
Light meal	**6 h**
Heavy meal (fat/meat)	**8 h**

ASA Physical Status Classification

- **I:** normal healthy (non-smoker, minimal alcohol).
- **II:** mild systemic disease, no functional limitation (smoker, social alcohol, **pregnancy, obesity 30–40, well-controlled DM/HTN, mild lung disease**).
- **III:** severe systemic disease with functional limitation (poorly controlled DM/HTN, COPD, **BMI ≥40, ESRD on dialysis, MI/CVA/stent >3 months ago**).
- **IV:** severe disease = constant threat to life (**MI/CVA/stent <3 months**, ongoing ischaemia, severe valve dysfunction, shock, sepsis, DIC).
- **V:** moribund, not expected to survive without operation (ruptured AAA, massive trauma, intracranial bleed).
- **VI:** brain-dead, organ donor.
- **E** suffix = emergency.

General anaesthesia

- **Definition:** drug-induced loss of consciousness, not arousable even by pain.
- **History:** nitrous oxide (Priestley 1772, Davy dental analgesia); **Morton — ether, Massachusetts General 1846 (successful)**; Wells (nitrous, 1845, discredited).
- **Components:** **immobility, amnesia, analgesia, loss of reflexes, lack of patient harm.**
- **Drugs:** hypnotics, neuromuscular blockers, analgesics.
- **Depth of sedation:** minimal (anxiolysis) → moderate (conscious) → deep → GA (no response to pain, airway intervention often needed).
- **Indications for GA:** neuromuscular blockade required, secure airway needed, not appropriate for MAC/regional, uncooperative/refusal, surgical pain not amenable to local.

Steps of GA

1. **Induction** (awake → anaesthetized): inhaled or IV.
- **Preoxygenation (denitrogenation):** 100% O₂ for 3 min → safety margin during apnoea.

- **Inhalational:** paediatric (no IV); **sevoflurane** (low pungency, high potency, rapid).
 - **IV** (commonest in adults): **propofol, thiopental, etomidate, ketamine, benzodiazepine-opioid.**
 - **Rapid Sequence Induction (RSI):** for **aspiration risk** (GERD, delayed emptying, unknown fasting, full stomach). Sequence: preoxygenate → hypnotic (propofol) → **rapid-onset NMB (succinylcholine/rocuronium)** → **cricoid pressure** → no mask ventilation → intubate → release cricoid after confirmation.
2. **Airway management:** mask ventilation → NMB → laryngoscopy/intubation or supraglottic.
3. **Maintenance:** **potent inhaled (halogenated) agents = mainstay** (titratable, ↓autonomic response, muscle relaxation); **nitrous oxide** (hypnosis + analgesia, but can't be sole — low potency); **propofol** (↓PONV, less laryngospasm); **TIVA (propofol + ketamine + opioid)** for evoked potential monitoring (e.g. scoliosis — inhaled agents affect signals).
4. **Recovery:** inhaled cleared via lungs; IV redistributed/metabolised/antidotes; **muscle relaxants reversed.**
5. **PACU:** recovery + direction for ongoing care.

High-yield one-liners

- **Fasting:** 2h clear liquids, 6h light meal, 8h heavy meal.
- **ASA:** I healthy → VI brain-dead; III/IV split by MI/CVA >3mo vs <3mo; E = emergency.
- **GA components:** immobility, amnesia, analgesia, loss of reflexes, no harm.
- **Preoxygenation** = 100% O₂ ×3min; sevoflurane for paediatric inhaled induction.
- **RSI for aspiration risk:** propofol + succinylcholine/rocuronium + cricoid pressure, no mask ventilation.
- **Maintenance:** halogenated agents mainstay; TIVA (propofol) for evoked potential monitoring; Morton 1846 ether.

PREOPERATIVE ASSESSMENT & GENERAL ANAESTHESIA (Revision Table)

Bold = high-yield. Amir M. Boujan.

FASTING & ASA

Topic	Key points
ASA fasting	Clear liquids 2h, breast milk 4h, formula 5–6h, non-human milk 6h, light meal 6h, heavy meal 8h
ASA I–II	I = healthy; II = mild disease (smoker, pregnancy, obesity 30–40, controlled DM/HTN)
ASA III–IV	III = severe + limitation (BMI≥40, ESRD dialysis, MI/stent >3mo); IV = constant threat (MI/stent <3mo , shock, sepsis)
ASA V–VI / E	V = moribund (ruptured AAA); VI = brain-dead donor; E = emergency suffix

GENERAL ANAESTHESIA

Topic	Key points
Definition	Drug-induced LOC, not arousable by pain
Components	Immobility, amnesia, analgesia, loss of reflexes, no harm
Depth	Minimal → moderate (conscious) → deep → GA (no pain response, airway intervention)
Indications	NMB needed, secure airway, not for MAC/regional, uncooperative, pain not amenable to local

STEPS OF GA

Step	Key points
Preoxygenation	100% O2 ×3 min (denitrogenation) — apnoea safety margin
Induction	IV (propofol/thiopental/etomidate/ketamine) commonest adult; inhaled (sevoflurane) paediatric
RSI	Aspiration risk → preoxygenate + propofol + succinylcholine/rocuronium + CRICOID PRESSURE + no mask ventilation → intubate
Airway	Mask → NMB → laryngoscopy/intubation or supraglottic
Maintenance	Halogenated agents = mainstay ; N2O (adjunct, low potency); propofol (↓PONV); TIVA for evoked potential monitoring (scoliosis)
Recovery/PACU	Inhaled via lungs; IV metabolised; NMB reversed; PACU monitoring